

**EXAME  
INGLÊS TÉCNICO**

Curso: LECC

Data: 04-08-2021

Turma: LECC11 e LECC12

Duração: 120 min.

Nome do Docente: Anastácio Macuácuá

Pontuação: 230

Nome do aluno \_\_\_\_\_

Importante: A fraude no exame de uma disciplina tem como consequência a reprovação na disciplina, sem possibilidade do infrator participar no exame de recorrência nem no exame especial (se existir) da disciplina em causa (alínea b, artigo 1 da ADENDA AO RPL).

**I. Reading.**

In 1952, a major computing company took a decision to get out of the business of making mainframe computers. They believed that there was only a market for four mainframes in the whole world. That company was IBM. The following year they reversed their decision.

In 1980, IBM decided that there was a market for 250,000 PCs, so they set up a special team to develop the first IBM PC. It went on sale in 1981 and set a world-wide standard for IBM-compatibility which, over the next ten years, was only seriously challenged by one other company, Apple Computers. Since then, over seventy million PCs made by IBM and other manufacturers have been sold. Over this period, PCs have become commodity items. Since IBM made the design non-proprietary, anyone can make them.

The history of the multi-billion dollar PC industry has been one of mistakes. Xerox Corporation funded the initial research on personal computers in their Palo Alto laboratory in California. However, the company failed to capitalize on this work, and the ideas that they put together went into the operating system developed for Apple's computers. This was a graphical interface: using a mouse, the user clicks on icons which represent the function to be performed.

The first IBM PC was developed using existing available electrical components. With IBM's badge on the box it became the standard machine for large corporations to purchase. When IBM were looking for an operating system, they went initially to Digital Research, who were market leaders in command-based operating systems (these are operating systems in which the users type in commands to perform a function). When the collaboration between IBM and Digital Research failed, IBM turned to Bill Gates, then

25 years old, to write their operating system.

Bill Gates founded Microsoft on the basis of the development of MS/DOS, the initial operating system for the IBM PC. Digital Research have continued to develop their operating system, DR/DOS, and it is considered by many people to be a better product than Microsoft's. However, without an endorsement from IBM, it has become a minor player in the market. Novell, the leaders in PC networking, now own Digital Research, so things may change.

The original IBM PC had a minimum of 16K of memory, but this could be upgraded to 512K if necessary, and ran with a processor speed of 4.77MHz. Ten years later, in 1991, IBM were making PCs with 16Mb of memory, expandable to 64Mb, running with a processor speed of 33MHz. The cost of buying the hardware has come down considerably as the machines have become commodity items. Large companies are considering running major applications on PCs, something which, ten years ago, no one would have believed possible of a PC. In contrast, many computers in people's homes are just used to play computer games.

The widespread availability of computers has in all probability changed the world for ever. The microchip technology which made the PC possible has put chips not only into computers, but also into washing-machines and cars. Some books may never be published in paper form, but may only be made available as part of public databases. Networks of computers are already being used to make information available on a world-wide scale.

1. Answer these questions about the text. (45pts)

1.1. What is the suitable title for the text?

1.2. How many mainframes did IBM think it was possible to sell in 1952?

1.3. How many PCs have now been sold?

1.4. Who paid for the initial research into PCs?

1.5. Which company later used the results of this research to develop their operating system?

1.6. What are command-based operating systems?

1.7. DR/DOS is an acronym. What does it stand for?

1.8. Since the invention of the IBM PC, many of its features have been improved. Which of the following features does the text *not* mention in this respect?

a) memory

b) speed

c) size

d) cost

1.9. Give three examples from the text of how the availability of computers has 'in all probability changed the world for ever'.

2. Using the line references given. Look back in the text and find words that have a similar meaning to: (40 pts)

2.1. international (lines 10-15)

2.2. contested (lines 15-20)

2.3. errors (lines 25-30)

2.4. paid for (lines 20-30)

2.5. buy (lines 45-50)

2.6. first (lines 60-65)

2.7. recommendation (lines 65-70)

2.8. improved (lines 75-80)



3. Read this passage about the structure of the processor and fill in the gaps using the words below. (45 pts)

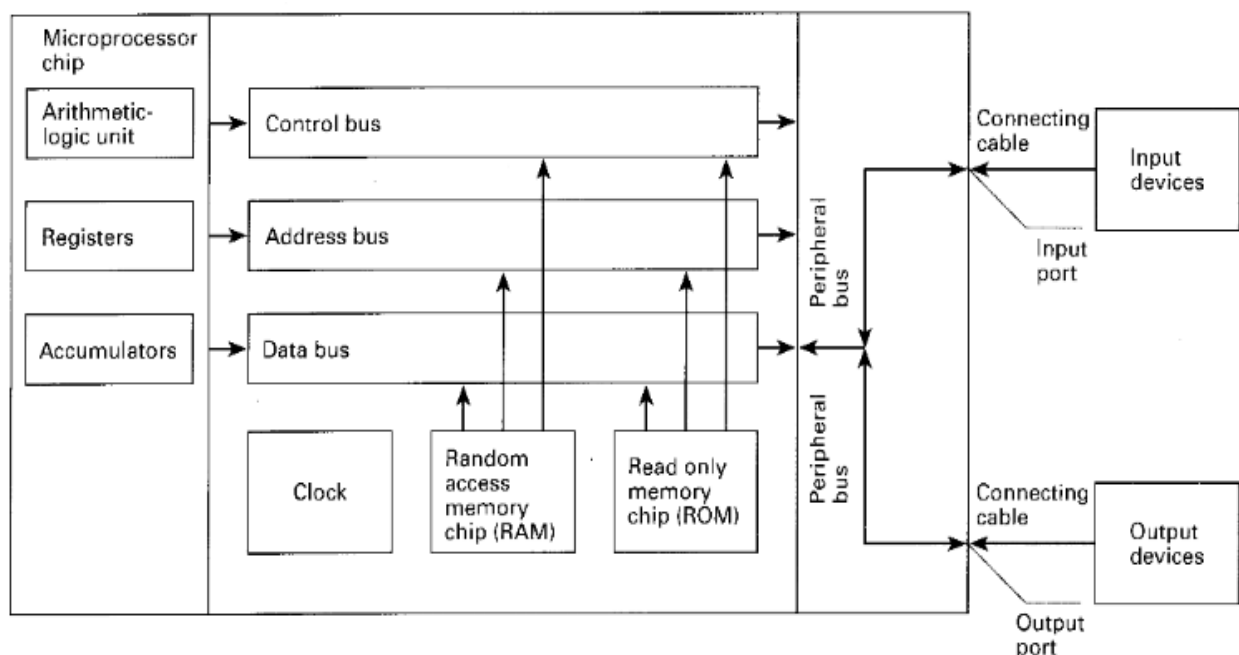
|                |              |                |                        |            |
|----------------|--------------|----------------|------------------------|------------|
| Adaptor boards | registers    | microprocessor | clock                  | conductive |
| buses          | system board | accumulators   | input or input devices |            |



### Structure of the processor

The processor consists of a \_\_\_\_1\_\_\_\_ which is a circuit board on which are mounted \_\_\_\_2\_\_\_\_ chips, memory chips, and other components linked together by \_\_\_\_3\_\_\_\_ lines or channels in the form of control, address, and data \_\_\_\_4\_\_\_\_. In addition, a processor has \_\_\_\_5\_\_\_\_, which are electronic circuits providing specialized functions such as graphics, or which a system board to \_\_\_\_6\_\_\_\_. The system board also consists of electronic devices, such as an electronic \_\_\_\_7\_\_\_\_ for controlling the speed of operation; \_\_\_\_8\_\_\_\_, which store numeric data during the course of processing; and various \_\_\_\_9\_\_\_\_, including sequence control register, address register, and function register.

4. Interpret the diagram below using appropriate language and process description. (30 pts)



5. Find the topic sentence, the supporting ideas, and arguments and examples by completing the lines after reading the paragraphs. You don't have to write every word in the main idea sentence or in each of the detail sentences.

**Paragraph 1 (15 pts)**

*The computer has changed the production of copy in the newspaper industry. There are three steps involved in the process: input, correction, and output. First, the computer numbers each story. counts words. and gives a listing of the length of each story. Then. a page is made up, advertisements are placed in. The copy is shifted or deleted. and corrections arc made. Finally, the computer hyphenates words, and the result of all this is a newspaper page.*

**Topic Sentence** \_\_\_\_\_

**Supporting idea 1** \_\_\_\_\_

**Supporting Idea 2** \_\_\_\_\_

**Argument 1** \_\_\_\_\_

**Argument 2** \_\_\_\_\_

**Paragraph 2 (15 pts)**

Railway companies use large computer systems to control ticket reservations and to give immediate information on the status of their trains. The computer system is connected by private telephone lines lo terminals in major train stations, and ticket reservations for customers arc made through these phone lines. The passenger's name, type of accommodation, and the train schedule is put into the computer's memory. On a typical day. a railway's computer system gets thousands of telephone calls about reservations, space on other railways. and requests for arrivals and departures. A big advantage of the railway computer ticket reservation system is its rapidity because a cancelled booking can be sold anywhere in the system ju.st a few seconds Later. Railway computer systems are not used for reservations alone. They are used for a variety of other jobs including train schedules. planning, freight and cargo loading, meal planning, personnel availability. accounting. and stock control.

**Topic Sentence** \_\_\_\_\_

**Supporting idea 1** \_\_\_\_\_

**Supporting Idea 2** \_\_\_\_\_

**Argument 1** \_\_\_\_\_

**Argument 2** \_\_\_\_\_

6. In no more than 250 words, write an **academic paragraph** talking about the cons and pros of machine translation. (40 pts)

Good Job